Chapter-4

Writing Research (Project) Proposal

What is research proposal?

A research proposal is a simply a structured, formal document that explains **what** you plan to research.

Main Components of Research Proposal

- Title
- Introduction
- Statement of the problem
- Significance of the research
- Scope and limitations of the study
- Review of literature
- Objectives of the research
- Operational definitions of terms used
- Hypothesis
- Methodology Used
- Time schedule/work plan
- Budget/estimated cost built up
- Organization of the report/chapter outline
- Bibliography/References
- Conclusions
- Appendix

Title

- First page – title of the research; name of institute/organization where the researcher belongs with logo, name of the supervisor, co-supervisor; date.

- In selecting the title, the following points should be taken care of:
  - ✓ Reflect the idea of the research
  - ✓ Be self-explanatory
  - ✓ Be brief
  - ✓ Language be simple and unambiguous
  - ✓ Be specific to a particular domain
  - ✓ Bracket be avoided
  - ✓ Avoid confusing meaning



ADDIS ABABA UNIVERSITY

ADDIS ABABA INSTITUTE OF TECHNOLOGY (AAiT)

SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING

DEPARTMENT OF CONTROL ENGINEERING

TITLE: SLIDING MODE CONTROL THROUGH FEEDBACK LINEARIZATIO OF AN INVERTED PENDULUM

BY: - [redacted] **(GSR/3369/11)**

Submitted to: Assoc. Prof. Mengesha Mamo (PhD)

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Addis Ababa

Ethiopia

Introduction

The main purpose of the introduction is to provide the necessary background or context for your research problem. How to frame the research problem is perhaps the biggest problem in proposal writing.

Statement of the problem

- Contains brief problem that needs to be addressed
- Statement of problem should be clear on the scope and feasible (validity of the hypothesis for engineering ,schedule, resource's)

Significance of the research

- Importance of the research
- Bridging the knowledge gaps
- Useful to the society/community

Scope and limitations of the study

- Boundaries of the research
- Aspects and issues addressed
- Shortcomings of the research – resource and time constraints

Review of literature

- Researches so conducted a valid and essential information
- Issues discussed; unaddressed issues
- Relevancy to the present research
- Finding out information/knowledge gaps

3. Research Process

Typical literature review format

Authors in reference [*1*] have demonstrated that permanent magnet synchronous motors can be controlled sensorless using back EMF method. In this method, the three phase inverter terminal voltages are measured and compared to generate a proper pattern of inverter switching. The authors have demonstrated that this method is effective for speed above 10% the rated voltage with improved error as speed increases. There are three problems with this method. One problem is that it can not be used in low speed (less than 10% base speed). The second problem is that it requires voltage sensor which is usually difficult and expensive. The third problem is that the method is used only for trapezoidal flux distributed permanent magnet motor. It is not effective for permanent magnet synchronous motor whose flux is distributed sinusoidal.

Authors in [*2*] have proposed sensorless control method using high frequency current injection to the motor. The injected current frequency is between the inverter carrier frequency and the fundamental frequency. The difficulty with this method is possible torque ripple due to the injected frequency current and the difficulty in filtering the carrier frequency component current as it is affected both by the carrier frequency and the fundamental current. …

Carrier frequency component current method (CFCCM) Sensorless control is effective in the whole speed range including zero speed as the rotor position is based on the inductance variation due to the magnetic saliency. It doesn't require any voltage sensor. It uses the current sensors which are required for current feedback anyway. It doesn't require any current injection, but utilize the redundant currier frequency component current as rotor position information source. ….

Objectives of the research

- Represents expected outcomes
- Concise, clear-cut, expressed in simple language, precise, self-explanatory
- Two types of objectives : General and Specific.
- The **specific objective** lead to the **general objectives** usually step wise

Example

➢**General objective**

The general objective of this work is to undergo **theoretical study**, **modeling** and **simulation** of an Inverted pendulum using SMC controller through input-out feedback linearization.

➢**Specific objective**

✓ To derivation and validation of a mathematical model of an inverted pendulum.

✓ To design and simulation of SMC controller.

✓ To design of Input-Output feedback linearization controller.

✓ To Experimental verification

Operational definitions of terms used

- Clear-cut meaning of the terms used

Hypothesis

Its an idea proposed to solve the problem

- Proposition subject to verification
- Guide/lead the research

Methodology Used

- Selection of appropriate approach
- Tools/techniques to be used
- Data collection techniques
- Data processing, analysis, interpretation techniques
- Data/information presentation techniques

Time schedule/work plan

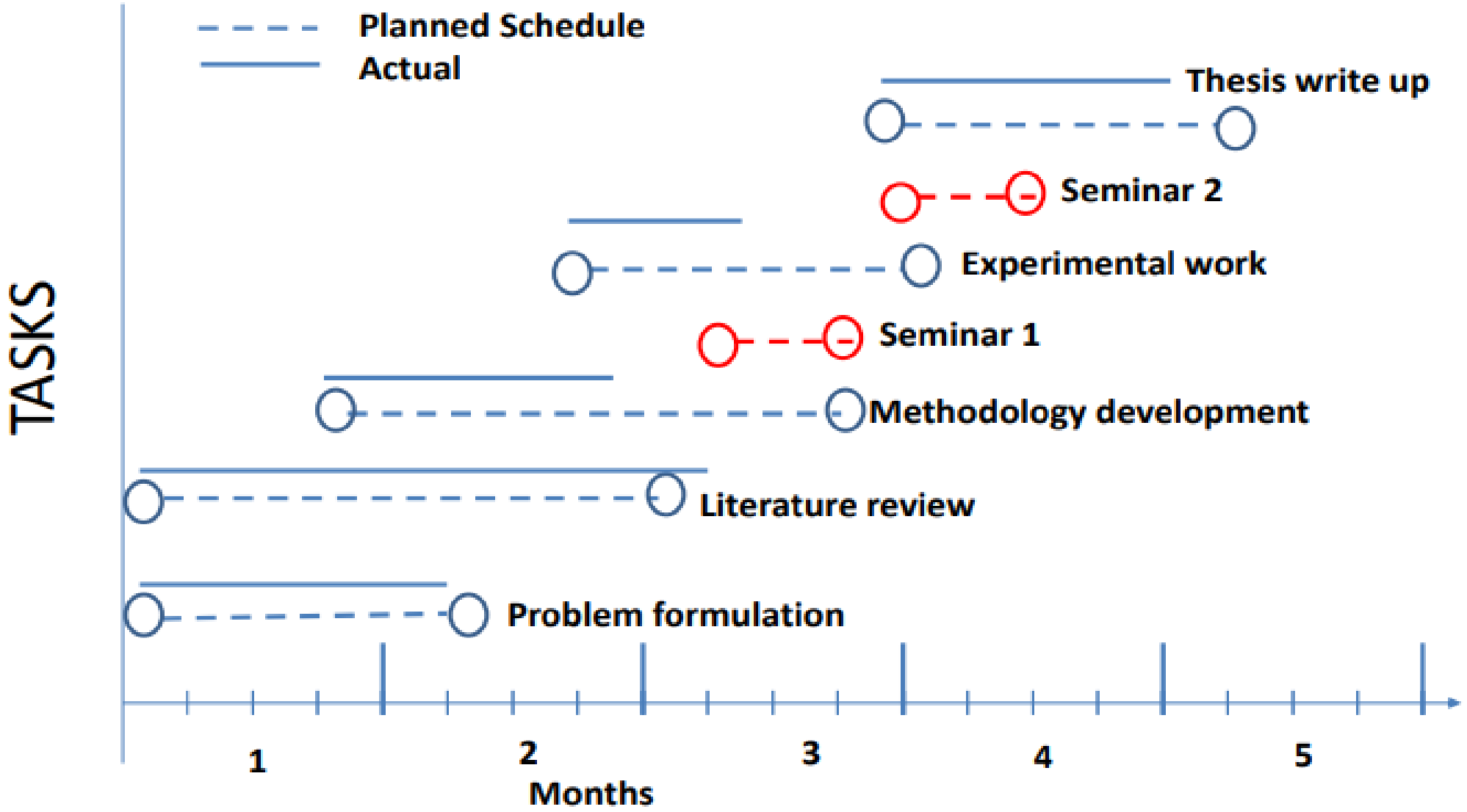
- Time line of the assigned tasks

Budget/estimated cost built up

- Cost built up – resource personnel, support staff, stationery, transport, utilities, house rents, miscellaneous, etc

Organization of the report/chapter outline

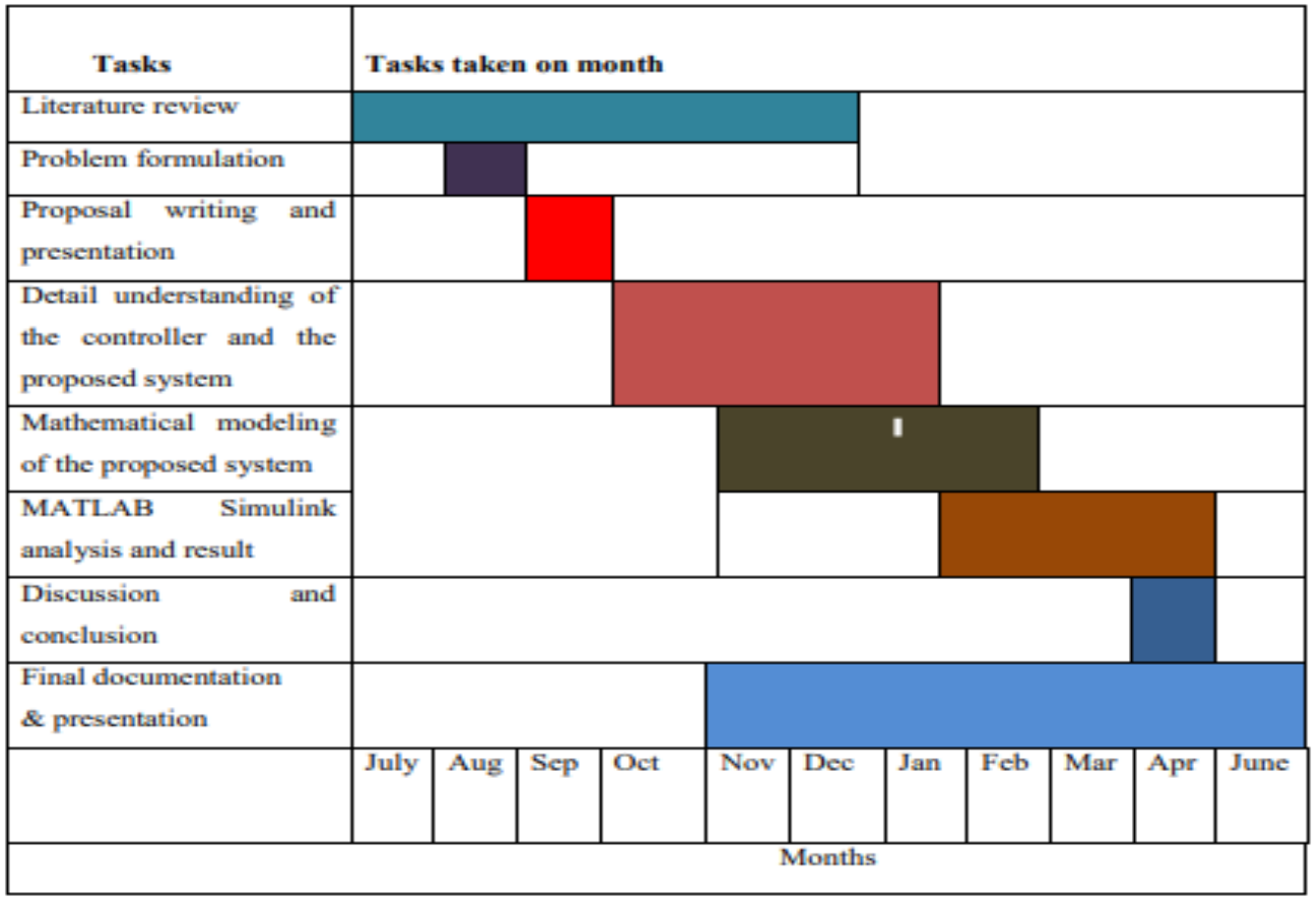
- Outline of the dissertation/report
- Five major parts of the report: **abstract, acknowledgement** , introduction , literature review, Methodology ,result and discussion, conclusion and recommendation.


Planned Schedule
Actual
Thesis write up
Seminar 2
Experimental work
Seminar 1
Methodology development
Literature review
Problem formulation
TASKS
1
2
3
4
5
Months

Example time schedule

4 Time schedule plan

The table below summarizes the overall work plan of the proposed work



Example on budget plan

5 Budget plan

The overall budget plan for the proposed thesis is summarized in thetable below

No	Tasks	Cost per unit (in birr)	Total cost (in birr)
1	Communication to exchange Information through mobile phone, internet…….	50	3,000
2	Pen & A4 paper for design calculation	150	1000
3	Proposal documentation	30	60
4	Buying books, journals and other relevant Papers as well as equipment	-	9,000
5	Transportation to learn and improve Software	1500	6,000
6	Final paper printing and Binding documentation	200	500
Subtotal in birr			**19,560**
7	Miscellaneous: If inflation occurs, 35% Cost could be added.	-	**6,846**
Total in birr			**26,406**

Bibliography/References

- Related documents to be consulted/studies
- Follow technicalities in writing bibliography/references
- To be presented in alphabetic order
- To be presented in classified manner viz., manuscripts, books, journals, commission reports, newspapers, etc

Example:

Knapik, J. J., Cosio-Lima, L. M., and Reynolds, K. L. (2015) 'Efficacy of functional movement screening for predicting injuries in coast guard cadets', *The Journal of Strength and Conditioning Research*, 29 (5), pp. 1157-1162.

References

"A reference list is the detailed list of references that are cited in your work".

- Banerjee, A. and Watson, T.F. (2011) *Pickard's manual of operative dentistry.* 9th edn. Oxford: Oxford University Press.
- Davidson, A. (2013) 'The Saudi Marathon Man', *The New Yorker,* 16 April. Available at: http://www.newyorker.com/news/daily-comment/the-saudi-marathon-man (Accessed: 22 June 2015).
- Hislop, V. (2014) *The sunrise.* Available at http://www.amazon.co.uk/kindlestore (Downloaded: 17 June 2015).
- Homer (1997) *The Iliad.* Translated by J. Davies. Introduction and notes by D. Wright. London: Dover Publications.

Example

References

[1] Mr. Khalil Sultan "Inverted Pendulum Analysis, Design and Implementation" IIEE Visionaries Document Version 1.0

[2] ASUK AMBA J. "Feedback linearization sliding mode and swing up control for the inverted pendulum on a cart" University of Manchester 2015.

[3] Liu Shuang and Fang Jian "Linear Quadratic Regulator Design an Inverted pendulum" IEEE International Symposium on Computer, Consumer and Control 2014

[4] Chandan Kumar, Santosh Lal, Nilanjan Patra ,Kaushik Halder, Motahar Reza "Optimal Controller Design for Inverted Pendulum System based on LQR method" International Conference on Advanced Communication Control and Computing Technologies (ICACCCT). 2012 IEEE

[5] Saqib Irfan, Adeel Mehmood,Muhammad Tayyab Razzaq, Jamshed Iqbal "Advanced sliding mode control techniques for Inverted Pendulum:Modelling and simulation" , Engineering Science and Technology, an International Journal 21 (2018) 753–759

[6] Jia-Jun Wang "Simulation studies of inverted pendulum based on PID controllers" 19 (2011) 440–449

[7] Ratchatin Chanchareon, Viboon Sangveraphunsiri, and Supavut Chantranuwathana "Tracking Control of an Inverted Pendulum Using Computed Feedback Linearization Technique" ,IEEE International Conference on Robotics, Automation and Mechatronics (RAM 2006) 7-9 June 2006, Bangkok, Thailand.

[8] Jean-Jacques E. Slotine Weiping Li "Feedback Linearization" in Applied nonlinear control 2nd, Ed pp. 207-271

Conclusion

- State about the feasibility of the proposed research/study
- Upcoming challenges
- Upcoming treats and difficulties
- Make a conclusive remark

DOs and DO NOTs

DO

✓Produce/prepare a professional looking proposal

✓Make it interesting

✓Make it informative, meaningful

✓Write easy way to read

✓Present content in a page

✓Use clear headings/sub-headings

✓Be concise, precise

✓Check spelling, grammar

✓Present in accurate/acceptable format

DO NOTs

✓Use no word which you do not understand

✓Use of difficult word unimpressive to the readers/supervisor/authority

Format for Writing proposal

- **Introduction**
  - Background (Basic and recent development knowledge of the research area)
  - Problem statement
  - General and specific objectives
  - Scope of the project
  - significance, impact, benefit of the project
- **Methodology**
  - Literature survey – How data is to be generated and availability of tools and equipment, Data analysis methods and tools availability, Work breakdown and structure
  - Relevance to previous work and expected new contribution
  - Limitations

Cont.

- **Schedule**
  - Tasks with small charts with Milestones and deadlines
- **Budget**
  - List of equipment and tools with availability, cost, etc – Materials – Transportation, …
- **References** – UpToDate and relevant

Format for Writing Report

- Title page
- Approved sheet
- Acknowledgement
- Table of Content
- List of Figures
- List of Tables

Main body of the report

- Introduction Section
- Problem statement
- Assumptions
- Operational definitions
- Variables
- Abstract
- Review of Literature
- Objective
- Scope
- Significance of the research

Methodology Section

- Research design
- Research approach
- Data collection tool
- Data analysis

Result Section

Study findings can be represented in graphical and non graphical form in simulation or demonstration .

Final Section

- Discussions
- Conclusion

Implication Section

- Limitations and Recommendations
- Future work
- Reference
- Appendix.

END of CHAPTER 4!!!